Lecture 6.1: From Question to Estimating Equation

AREC 570 / ECON 530 · Unit 6

Lauren Chenarides

Before class

Assigned

- Angrist and Pischke, *Mostly Harmless Econometrics*, Chapters 1 to 3 (Angrist and Pischke 2009)
- Jurajda, “Lecture Notes on Identification Strategies” (Jurajda 2007)
- Timmins and Schlenker, “Reduced-Form Versus Structural Modeling in Environmental and Resource Economics” (Timmins and Schlenker 2009)

Also used in these slides

- Bellemare, “How to Write Applied Papers in Economics,” from Lecture 2.3 (Bellemare 2020)
- AJAE Author Guidelines (Agricultural and Applied Economics Association, n.d.)
- Athey and Imbens, “The State of Applied Econometrics,” on robustness and sensitivity (assigned for 6.3) (Athey and Imbens 2017)
- Gentzkow and Shapiro on keys, from Lecture 5.4 (Gentzkow and Shapiro 2014)
- Published example: Dalheimer and Bellemare (2026)
- Climate example: Mendelsohn et al. (1994); Schlenker et al. (2005); Deschênes and Greenstone (2007); Schlenker and Roberts (2009); Burke and Emerick (2016)

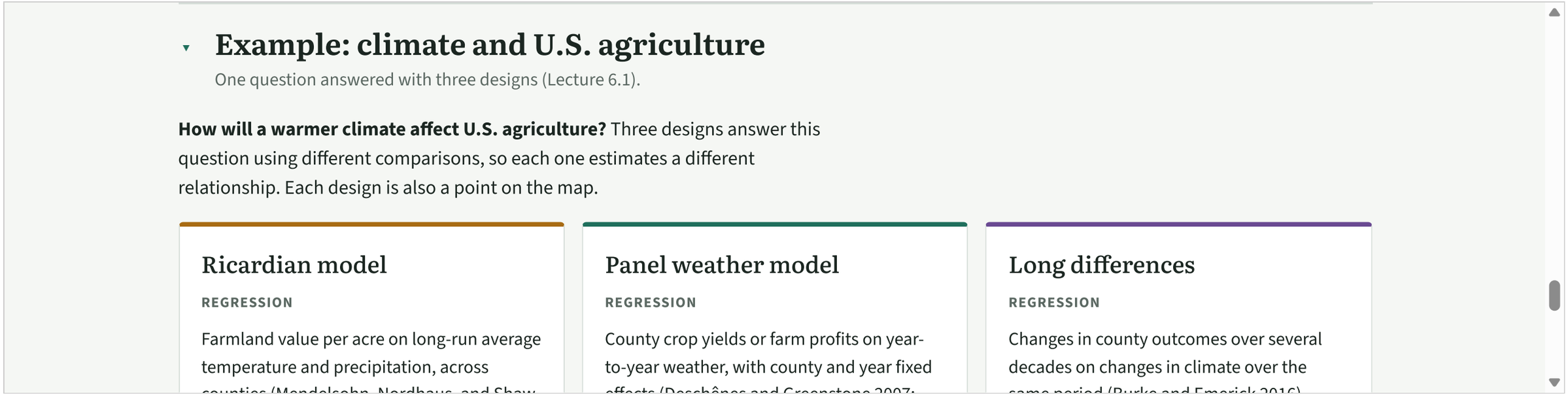
Readings and course handouts are available on Canvas.

The empirical strategy checklist lists what your section must report

- One page, parallel to the data checklist from Lecture 5.5
- Rows 1 to 9 cover what every paper reports; rows 10 to 17 depend on the design
- Each row is tagged with the lecture that covers it; today covers rows 1 to 9
- Built from Bellemare's Section 5, Angrist and Pischke's Chapter 1, and the AJAE Author Guidelines

Open the empirical strategy checklist

The methods map helps you match a method to your question



A map of methods, a decision tree, the four choices every empirical strategy states, structural estimation details for 6.2, and the climate example we use today. Use it as a reference while writing.

Open the methods map in a new tab

What must the empirical strategy section accomplish?

But first, where the empirical strategy section lives in your paper

1. Introduction
2. Background or Literature Review
3. Conceptual Framework
4. Data and Summary Evidence
5. **Empirical Strategy** & **Identification Strategy**
6. Results
7. Discussion (or Policy Implications, or Managerial Implications)
8. Conclusion

Bellemare calls this section the Empirical Framework, and some papers place it before the data section (Bellemare 2020).

The empirical strategy section turns a prediction from your framework into an estimating equation

- Conceptual framework (Unit 4) gives you guidance on developing testable predictions
- Data (Unit 5) gives you guidance on describing your data, the unit of observation, and the measures that make your data fit for use
- Methods (this unit) explains how to turn the conceptual model into an equation with estimable parameters, and why the data identify them
- A referee should be able to judge the design before seeing a result

Four terms come up in every empirical strategy section

- **Parameter of interest:** the coefficient that answers your research question
- **Estimator:** the rule that turns data into an estimate of that parameter, such as OLS, maximum likelihood, or a difference in means
- **Identifying assumption:** the condition under which the parameter you estimate equals the relationship you want to learn about
- **Inference:** how you measure the uncertainty in the estimate, through standard errors, clustering, and hypothesis tests

Bellemare splits the section into what you estimate and why it identifies a causal effect

Estimation strategy

The equations, the estimator, and how inference is done

Identification strategy

The feature of the data that supports a causal interpretation, and what threatens it

Use these as the two parts of your section (Bellemare 2020).

A published paper states the equation, then defines every term

2.1 Estimation Strategy

We estimate the relationship between the extent of participation in GAVCs by a country in a given year and (i) food price levels as well as (ii) volatility in the same country-year. To do so, we estimate the following baseline equation:

$$\Delta p_{it} = \beta_1 \Delta GAVC_{it} + \gamma_1' \Delta X_{it} + \eta_{1t} + e_{1it}, \quad (1)$$

where p_{it} is the real consumer price level for food in county i in year t, $GAVC_{it}$ is the extent of GAVC participation by the same country in the same year, and X_{it} is a vector of control variables that includes the time-variant country-level characteristics listed in Appendix Table 5. We also include year fixed effects η_t to control for shocks affecting all countries in each given year. Lastly, e_{it} is an error term with mean zero.

In this case, the parameter of interest is β_1 which, in Equation 1, captures the association between participation in GAVCs and the real food price level. We estimate Equation 1

Source: Dalheimer and Bellemare (2026), Section 2.1 (accepted version posted on Bellemare's website).

The identification strategy explains what in the data supports a causal interpretation

2.2 Identification Strategy

When analyzing the impact of trade on food prices, endogeneity poses a significant challenge. One issue could be reverse causation, where changes in food prices influence trade patterns instead of being solely driven by them. For instance, rising food prices in a country might lead to higher imports to meet demand, creating a feedback loop between trade and prices that complicates causal inference.

Omitted variables present another problem, as external factors like extreme weather events, economic policies, or political instability can simultaneously affect both trade volumes and food prices. For example, a natural disaster in an exporting region may disrupt trade while also driving up global prices. Ignoring such variables could result in misleading estimates of the trade-price relationship.

Finally, data limitations and structural market features can introduce additional biases. Inaccuracies in trade or price data, along with complexities such as market concentration

Source: Dalheimer and Bellemare (2026), Section 2.2 (accepted version posted on Bellemare's website).

Six questions organize the section, in the order they appear in a paper

Subsection	Question	Where the answer appears
Estimation strategy	1. What is the relationship of interest?	The estimating equation and the parameter of interest
	2. How will you measure uncertainty?	The estimator, the standard errors, the clustering level, and why
Identification strategy	3. Is that relationship causal?	Whether the parameter is causal, and if not, what it describes
	4. What experiment would ideally measure it?	The effect you are trying to estimate
	5. What variation identifies it, and what must be true of that variation?	The identifying assumption
	6. What could make the identifying assumption fail?	The threats, and the design choice or robustness check that handles each

Adapted from Angrist and Pischke (2009), Chapter 1, and Bellemare (2020), Section 5.

What’s included in the estimation strategy subsection

Write the equation so a referee never has to ask what a symbol means

- Latin letters for variables, Greek letters for parameters
- Subscripts run from the smallest unit to the largest
- Define every term directly below the equation
- Give each specification its own coefficient label
- The unit of observation matches the key in your data section

Source: Bellemare (2020), Section 5.1.

The unit of observation describes a row; the key identifies it in the file

	Unit of observation	Key
What it is	What one row represents, in words	The field or fields whose values are unique for every row
Panel weather model	A county in a year	County FIPS code and year
Price-gouging data (Lecture 5.5)	One product at one store in one week	Store, product, and week identifiers
Where it appears	The data section and the equation's subscripts	The code that builds and merges the data

The unit tells the reader what each row means. The key lets you check that there is exactly one row per unit before you estimate (Gentzkow and Shapiro 2014).

The estimation strategy also names the estimator and says how inference is done

- The estimator by name, such as OLS, a linear probability model, probit, or maximum likelihood, and why
- The hypothesis being tested
- What the standard errors are robust to, such as heteroskedasticity; “robust standard errors” alone is not enough
- Whether and how standard errors are clustered, at what level, and why
- Whether sampling weights are used, and how they were constructed

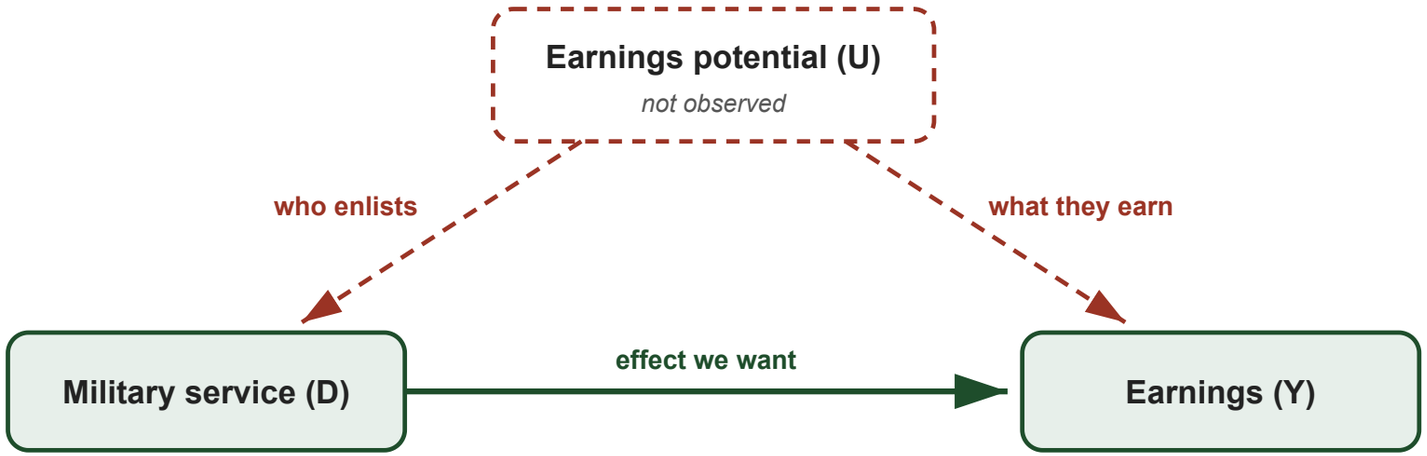
Source: Bellemare (2020), Section 5.1.

What’s included in the identification strategy subsection

Identification asks whether your data and assumptions determine the parameter you want to estimate

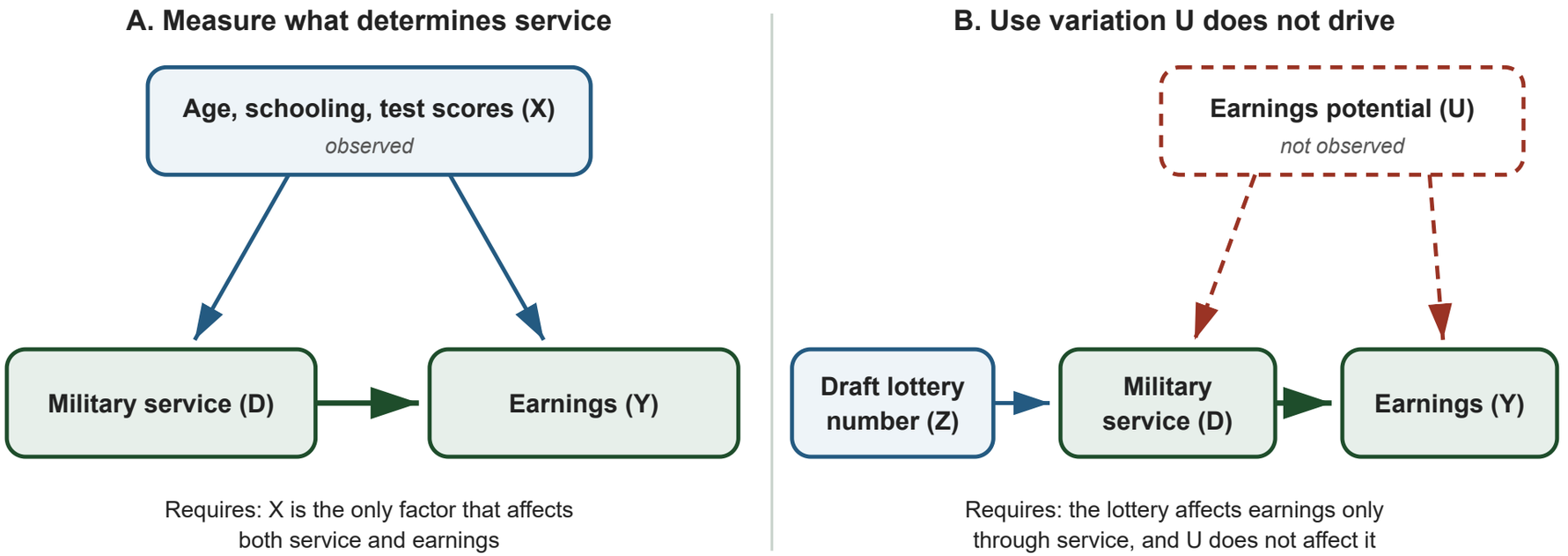
- Suppose you had data on the entire population. The parameter is identified if only one value of it is consistent with those data
- If more than one value fits, the parameter is not identified, and a larger sample or a different estimator will not fix it
- Whether a parameter is identified depends on assumptions about how the data were generated
- A directed acyclic graph (DAG) draws those assumptions: each arrow is a causal effect, and each missing arrow is an assumption
- **Example:** a paper wants to estimate the effect of military service on later earnings (Jurajda 2007, Example 1.1)

When an unobserved factor drives both treatment and outcome, the effect is not identified



A comparison of veterans and nonveterans picks up both the effect and the path through earnings potential. With unlimited data it still cannot separate them (Jurajda 2007, Example 1.1).

The effect is identified if you measure the common causes or use variation they do not drive



A: Angrist (1998), discussed in Angrist and Pischke (2009), Chapter 3. B: the Vietnam-era draft lottery, Angrist (1990), in Jurajda (2007), Example 3.11.

The identifying assumption says why the groups you compare are comparable

- Comparing treated and untreated units measures the treatment effect plus any difference that existed between the groups before treatment
- Jurajda's example: veterans may earn more because of military service, or because people with higher earnings potential were more likely to enlist
- That pre-existing difference is selection bias
- The identifying assumption states why selection bias is zero for your comparison
- Write it in one sentence, without notation

Bellemare asks every paper to address four threats to identification

Threat	What it means	Common responses
Reverse causality	The outcome affects the treatment	Measure treatment before the outcome; use an instrument
Unobserved heterogeneity	An omitted factor affects both treatment and outcome	Fixed effects, controls, or a design that holds the factor constant
Measurement error	The treatment is recorded with error; classical error biases the estimate toward zero	A second measure; an instrument
Spillovers	One unit's treatment affects another unit's outcome, now or later	Define units so spillovers stay within them; test for them

Source: Bellemare (2020), Section 5.2.

A threat to identification biases the estimate; a problem with inference misstates its precision

- **Threat to identification:** the estimate is centered on the wrong value, and a larger sample does not fix it
- **Problem with inference:** the estimate may be centered correctly, but the standard errors are too small, so confidence intervals are too narrow and tests reject too often
- **Panel weather example:** neighboring counties share weather shocks, so treating their errors as independent overstates precision; cluster at a level that contains that correlation and say why
- Either problem leads a referee to doubt the result, so the section addresses both

Source: Bellemare (2020), Sections 5.1 and 5.2.

Robustness checks vary your choices; sensitivity analyses relax your assumptions

- **Robustness check:** re-estimate with another defensible outcome measure, treatment measure, estimator, functional form, or sample, and show the conclusion holds (Bellemare 2020, sec. 6.2)
- **Sensitivity analysis:** ask how far the identifying assumption would have to fail before the conclusion changes, for example how strong an omitted variable would have to be (Athey and Imbens 2017)
- **Placebo test:** estimate an effect that should be zero; Bellemare counts these as robustness checks, and Athey and Imbens list them separately
- **Why papers include them:** researchers try many specifications before choosing one, so readers need evidence that the result was not selected for (Athey and Imbens 2017)

A robustness check asks whether your choices matter. A sensitivity analysis asks how wrong your assumption could be.

Example: climate and U.S. agriculture

The ideal experiment tells you which effect you are after

Question: How will a warmer climate affect U.S. agriculture?

- Ideal experiment: give some counties a permanently warmer climate, then wait decades
- Farmers would change crops, irrigation, and planting dates
- The target is a long-run effect that includes this adaptation
- A single hot year is a different experiment with a different answer

Three designs approximate this experiment, and each answers the six questions differently

Question	Ricardian	Panel weather	Long differences
1. What is the relationship of interest?			
2. How will you measure uncertainty?			
3. Is that relationship causal?			
4. What experiment would ideally measure it?			
5. What variation identifies it, and what must be true of that variation?			
6. What could make the identifying assumption fail?			

We fill in one column at a time. Each design responds to a weakness in the one before it.

The Ricardian model compares farmland values across counties with different climates

$$V_c = \alpha + f(T_c, P_c) + X_c^{'}\gamma + \varepsilon_c$$

- V_c: farmland value per acre in county c
- T_c, P_c: long-run average temperature and precipitation
- $f(\cdot)$: a quadratic in temperature and precipitation; its slope in T_c is the parameter of interest
- X_c: soil quality and other county characteristics

1. Relationship of interest: the long-run effect of climate on land values, including adaptation (Mendelsohn et al. 1994)

The Ricardian assumption fails if anything that varies with climate is omitted, and irrigation does

- **2. Uncertainty:** neighboring counties share soils, markets, and climate, so their errors are correlated
- **3. Causal?** Yes, if the identifying assumption holds
- **4. Experiment it approximates:** counties assigned different climates long ago
- **5. Variation and assumption:** differences in climate across counties; no omitted factor varies with climate and affects land values
- **6. What could make it fail:** irrigated farms in drier, hotter counties draw water from elsewhere and have higher land values; Schlenker, Hanemann, and Fisher re-estimate the model for dryland counties (Schlenker et al. 2005)

The panel model removes fixed county differences by comparing each county with itself across years

$$y_{ct} = g(W_{ct}) + \gamma_c + \delta_t + \varepsilon_{ct}$$

- y_{ct}: log crop yield or farm profit in county c, year t
- W_{ct}: growing-season weather, such as degree days and precipitation
- $g(\cdot)$: the relationship between weather and the outcome; its slope is the parameter of interest
- γ_c: county fixed effects; δ_t: year fixed effects

1. Relationship of interest: the short-run response of yields or profits to weather (Deschênes and Greenstone 2007; Schlenker and Roberts 2009)

Fixed effects solve the irrigation problem and change the relationship estimated

- **2. Uncertainty:** weather shocks hit neighboring counties together, so their errors are correlated
- **3. Causal?** Yes, for the effect of weather, if the identifying assumption holds
- **4. Experiment it approximates:** a randomly assigned hot or cool year
- **5. Variation and assumption:** year-to-year deviations in weather within a county; those deviations are as good as random
- **6. What could make it fail:** measurement error in gridded weather, and price changes that spill over across counties (a later slide covers all four)

Farmers cannot fully adapt to one hot year, so the long-run effect can be larger or smaller than this estimate (Timmins and Schlenker 2009).

Long differences compare slow changes in climate over decades

$$\Delta\bar{y}_c = \alpha_s + \beta\,\Delta\bar{W}_c + \Delta\bar{\varepsilon}_c$$

- $\Delta\bar{y}_c$: change in county c's multi-year average outcome between two periods
- $\Delta\bar{W}_c$: change in its multi-year average weather over the same periods
- α_s: state fixed effects; β: the parameter of interest

1. Relationship of interest: the response to gradual climate change, including adaptation (Burke and Emerick 2016)

Long differences recover the adaptation that the panel model misses

- **2. Uncertainty:** one observation per county leaves far fewer observations than the panel, and errors are still correlated across neighboring counties
- **3. Causal?** Yes, for the effect of gradual change, if the identifying assumption holds
- **4. Experiment it approximates:** a climate that warms slowly over decades
- **5. Variation and assumption:** differences in climate trends across counties within a state; those trends are unrelated to other trends in outcomes
- **6. What could make it fail:** other trends, such as technology adoption, that move with climate trends

Each design answers the six questions differently

Question	Ricardian	Panel weather	Long differences
1. What is the relationship of interest?	Long-run effect on land values	Short-run response to weather	Response to gradual change
2. How will you measure uncertainty?	Errors correlated across space	Errors correlated across space	Fewer observations; errors correlated across space
3. Is that relationship causal?	Yes, if the identifying assumption holds	Yes, if the identifying assumption holds	Yes, if the identifying assumption holds
4. What experiment would ideally measure it?	Different climates long ago	A random hot or cool year	A slowly warming climate
5. What variation identifies it, and what must be true of that variation?	Across counties; no omitted factor varies with climate	Within a county across years; weather deviations as good as random	Trends within a state; unrelated to other outcome trends
6. What could make the identifying assumption fail?	Omitted factors such as irrigation	Measurement error; price spillovers	Other trends that move with climate

In the panel weather model, each threat has an answer

Threat	In the panel weather model
Reverse causality	Farm output does not change the weather
Unobserved heterogeneity	County effects absorb fixed differences; year effects absorb national shocks
Measurement error	Gridded weather misses farm conditions; check a second weather data set
Spillovers	Year effects absorb national price changes, so the estimate excludes them

This applies the four threats to one design. Each row needs an answer: the threat does not apply, the design handles it, or a robustness check tests it.

Writing the section

Put the six questions in your draft as comments, then answer each in prose

```latex
\section{Empirical Strategy}

\subsection{Estimation Strategy}
% What is the relationship of interest?
%   equation, every term defined, parameter of interest
% How will you measure uncertainty?
%   standard errors, clustering level, and why

\subsection{Identification Strategy}
% Is that relationship causal?
% What experiment would ideally measure it?
% What variation identifies it, and what must be true of that variation?
% What could make the identifying assumption fail?
%   each threat, and the design choice or robustness check that handles it
```

Lines that start with `%` do not appear in the compiled PDF. In Word, use comments; in Quarto or Markdown, use `<!-- -->`.

Writing task: your estimating equation and identifying assumption

For your own proposal:

1. Write the estimating equation
2. Define every term, including subscripts and the unit of observation
3. Name the parameter of interest and say whether it is causal
4. State the identifying assumption in one sentence

References

Agricultural and Applied Economics Association. n.d. *American Journal of Agricultural Economics: Author Guidelines*. Oxford University Press. Accessed October 27, 2026. https://academic.oup.com/ajae/pages/Author_Guidelines.

Angrist, Joshua D., and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.

Athey, Susan, and Guido W. Imbens. 2017. "The State of Applied Econometrics: Causality and Policy Evaluation." *Journal of Economic Perspectives* 31 (2): 3–32. https://doi.org/10.1257/jep.31.2.3.

Bellemare, Marc F. 2020. *How to Write Applied Papers in Economics*. Working paper, University of Minnesota.

Burke, Marshall, and Kyle Emerick. 2016. "Adaptation to Climate Change: Evidence from US Agriculture." *American Economic Journal: Economic Policy* 8 (3): 106–40. https://doi.org/10.1257/pol.20130025.

Dalheimer, Bernhard, and Marc F. Bellemare. 2026. "Global Agricultural Value Chains and Food Prices." *American Journal of Agricultural Economics* 108 (5): 1282–99. https://doi.org/10.1111/ajae.70030.

Deschênes, Olivier, and Michael Greenstone. 2007. "The Economic Impacts of Climate Change: Evidence from Agricultural Output and Random Fluctuations in Weather." *American Economic Review* 97 (1): 354–85. https://doi.org/10.1257/aer.97.1.354.

Deschênes, Olivier, and Michael Greenstone. 2012. "The Economic Impacts of Climate Change: Evidence from Agricultural Output and Random Fluctuations in Weather: Reply." *American Economic Review* 102 (7): 3761–73. https://doi.org/10.1257/aer.102.7.3761.

Fisher, Anthony C., W. Michael Hanemann, Michael J. Roberts, and Wolfram Schlenker. 2012. "The Economic Impacts of Climate Change: Evidence from Agricultural Output and Random Fluctuations in Weather: Comment." *American Economic Review* 102 (7): 3749–60. https://doi.org/10.1257/aer.102.7.3749.

Gentzkow, Matthew, and Jesse M. Shapiro. 2014. *Code and Data for the Social Sciences: A Practitioner's Guide*. University of Chicago mimeo. https://web.stanford.edu/~gentzkow/research/CodeAndData.pdf.

Jurajda, Štěpán. 2007. *Lecture Notes on Identification Strategies*. Lecture notes.

Mendelsohn, Robert, William D. Nordhaus, and Daigee Shaw. 1994. “The Impact of Global Warming on Agriculture: A Ricardian Analysis.” *American Economic Review* 84 (4): 753–71.

Schlenker, Wolfram, W. Michael Hanemann, and Anthony C. Fisher. 2005. “Will U.S. Agriculture Really Benefit from Global Warming? Accounting for Irrigation in the Hedonic Approach.” *American Economic Review* 95 (1): 395–406. https://doi.org/10.1257/0002828053828455.

Schlenker, Wolfram, and Michael J. Roberts. 2009. “Nonlinear Temperature Effects Indicate Severe Damages to U.S. Crop Yields Under Climate Change.” *Proceedings of the National Academy of Sciences* 106 (37): 15594–98. https://doi.org/10.1073/pnas.0906865106.

Timmins, Christopher, and Wolfram Schlenker. 2009. “Reduced-Form Versus Structural Modeling in Environmental and Resource Economics.” *Annual Review of Resource Economics* 1: 351–80. https://doi.org/10.1146/annurev.resource.050708.144119.

Course navigation

- All lectures
- Schedule and preparation
- Course reading list
- Empirical strategy checklist
- Methods map
- Download this lecture as a PDF